

Name-Vishesh Singh Bhadauria

Email id- Vishesh.Bhadauria@Coforge.com

Emp id- 85011029

ISTQB – MCQ's

1 When what is visible to end-users is a deviation from the specific or expected behavior, this is called:

- a) an error
- b) a fault
- c) a failure
- d) a defect
- e) a mistake

2 Regression testing should be performed:

- v) every week
- w) after the software has changed
- x) as often as possible
- y) when the environment has changed
- z) when the project manager says
- a) v & w are true, x – z are false
- b) w, x & y are true, v & z are false
- c) w & y are true, v, x & z are false
- d) w is true, v, x y and z are false
- e) all of the above are true

3 IEEE 829 test plan documentation standard contains all of the following except: a) test items

- b) test deliverables
- c) test tasks
- d) test environment
- e) test specification

4 Testing should be stopped when:

- a) all the planned tests have been run
- b) time has run out
- c) all faults have been fixed correctly
- d) both a) and c)
- e) it depends on the risks for the system being tested

5 Order numbers on a stock control system can range between 10000 and 99999 inclusive. Which of the following inputs might be a result of designing tests for only valid equivalence classes and valid boundaries:

- a) 1000, 5000, 99999
- b) 9999, 50000, 100000
- c) 10000, 50000, 99999
- d) 10000, 99999

6 Consider the following statements about early test design:

- i. early test design can prevent fault multiplication
- ii. faults found during early test design are more expensive to fix
- iii. early test design can find faults
- iv. early test design can cause changes to the requirements
- v. early test design takes more effort

a) i, iii & iv are true. ii & v are false

b) iii is true, i, ii, iv & v are false

c) iii & iv are true. i, ii & v are false

d) i, iii, iv & v are true, ii is false

e) i & iii are true, ii, iv & v are false

7 Non-functional system testing includes:

a) testing to see where the system does not function properly

b) testing quality attributes of the system including performance and usability

c) testing a system feature using only the software required for that action d) testing a system feature using only the software required for that function e) testing for functions that should not exist

8 Which of the following is NOT part of configuration

management: a) status accounting of configuration items

b) auditing conformance to ISO9001

c) identification of test versions

d) record of changes to documentation over time

e) controlled library access

9 Which of the following is the main purpose of the integration strategy

for integration testing in the small?

a) to ensure that all of the small modules are tested adequately

b) to ensure that the system interfaces to other systems and networks

c) to specify which modules to combine when and how many at once

d) to ensure that the integration testing can be performed by a small

team e) to specify how the software should be divided into modules

10 What is the purpose of test completion criteria in a test plan:

a) to know when a specific test has finished its execution

b) to ensure that the test case specification is complete

c) to set the criteria used in generating test inputs

d) to know when test planning is complete

e) to plan when to stop testing

11 Consider the following statements

i. an incident may be closed without being fixed

ii. incidents may not be raised against documentation

iii. the final stage of incident tracking is fixing

iv. the incident record does not include information on test environments v. incidents should be raised when someone other than

the author of the software performs the test

a) ii and v are true, i, iii and iv are false

b) i and v are true, ii, iii and iv are false

c) i, iv and v are true, ii and iii are false

d) i and ii are true, iii, iv and v are false

e) i is true, ii, iii, iv and v are false

12 Given the following code, which is true about the minimum number of test cases required for full statement and branch coverage:

Read P

Read Q

IF P+Q > 100 THEN

Print "Large"

ENDIF

If P > 50 THEN

Print "P Large"

ENDIF

a) 1 test for statement coverage, 3 for branch coverage

b) 1 test for statement coverage, 2 for branch coverage

c) 1 test for statement coverage, 1 for branch coverage

d) 2 tests for statement coverage, 3 for branch coverage

e) 2 tests for statement coverage, 2 for branch coverage

13 Given the following:

Switch PC on

Start "outlook"

IF outlook appears THEN

Send an email

Close outlook

a) 1 test for statement coverage, 1 for branch coverage

b) 1 test for statement coverage, 2 for branch coverage

c) 1 test for statement coverage. 3 for branch coverage

d) 2 tests for statement coverage, 2 for branch coverage

e) 2 tests for statement coverage, 3 for branch coverage

14 Given the following code, which is true:

IF A > B THEN

C = A – B

ELSE

C = A + B

ENDIF

Read D

IF C = D Then

Print "Error"

ENDIF

a) 1 test for statement coverage, 3 for branch coverage

b) 2 tests for statement coverage, 2 for branch coverage

c) 2 tests for statement coverage. 3 for branch coverage

d) 3 tests for statement coverage, 3 for branch coverage

e) 3 tests for statement coverage, 2 for branch coverage

15 Consider the following:

Pick up and read the newspaper

Look at what is on television

If there is a program that you are interested in watching then switch the television on and watch the program

Otherwise

Continue reading the newspaper

If there is a crossword in the newspaper then try and complete the

crossword a) SC = 1 and DC = 1

b) SC = 1 and DC = 2

c) SC = 1 and DC = 3

d) SC = 2 and DC = 2

e) SC = 2 and DC = 3

16 The place to start if you want a (new) test tool is:

a) Attend a tool exhibition

b) Invite a vendor to give a demo

c) Analyse your needs and requirements

d) Find out what your budget would be for the tool

e) Search the internet

17 When a new testing tool is purchased, it should be used first by:

a) A small team to establish the best way to use the tool

b) Everyone who may eventually have some use for the tool

c) The independent testing team

d) The managers to see what projects it should be used in

e) The vendor contractor to write the initial scripts

18 What can static analysis NOT find?

a) The use of a variable before it has been defined

b) Unreachable ("dead") code

c) Whether the value stored in a variable is correct

d) The re-definition of a variable before it has been used

e) Array bound violations

19 Which of the following is NOT a black box technique:

a) Equivalence partitioning

b) State transition testing

c) LCSAJ

d) Syntax testing

e) Boundary value analysis

20 Beta testing is:

a) Performed by customers at their own site

b) Performed by customers at their software developer's site

c) Performed by an independent test team

d) Useful to test bespoke software

e) Performed as early as possible in the lifecycle

21 Given the following types of tool, which tools would typically be used by developers and which by an independent test team:

- i. static analysis
- ii. performance testing
- iii. test management
- iv. dynamic analysis
- v. test running
- vi. test data preparation

- a) developers would typically use i, iv and vi; test team ii, iii and v
- b) developers would typically use i and iv; test team ii, iii, v and vi
- c) developers would typically use i, ii, iii and iv; test team v and vi
- d) developers would typically use ii, iv and vi; test team I, ii and v
- e) developers would typically use i, iii, iv and v; test team ii and vi

22 The main focus of acceptance testing is:

- a) finding faults in the system
- b) ensuring that the system is acceptable to all users
- c) testing the system with other systems
- d) testing for a business perspective
- e) testing by an independent test team

23 Which of the following statements about the component testing standard is false:

- a) black box design techniques all have an associated measurement technique
 - b) white box design techniques all have an associated measurement technique
 - c) cyclomatic complexity is not a test measurement technique
 - d) black box measurement techniques all have an associated test design technique
 - e) white box measurement techniques all have an associated test design technique
- 24 Which of the following statements is NOT true:

- a) inspection is the most formal review process
- b) inspections should be led by a trained leader
- c) managers can perform inspections on management documents
- d) inspection is appropriate even when there are no written documents
- e) inspection compares documents with predecessor (source) documents

A typical commercial test execution tool would be able to perform all of the following EXCEPT:

- a) generating expected outputs
- b) replaying inputs according to a programmed script
- c) comparison of expected outcomes with actual outcomes
- d) recording test inputs
- e) reading test values from a data file

26 The difference between re-testing and regression testing is

- a) re-testing is running a test again; regression testing looks for unexpected side effects
- b) re-testing looks for unexpected side effects; regression testing is repeating those tests
- c) re-testing is done after faults are fixed; regression testing is done earlier
- d) re-testing uses different environments, regression testing

uses the same environment

e) re-testing is done by developers, regression testing is done by independent testers

27 Expected results are:

a) only important in system testing

b) only used in component testing

c) never specified in advance

d) most useful when specified in advance

e) derived from the code

28 Test managers should not:

a) report on deviations from the project plan

b) sign the system off for release

c) re-allocate resource to meet original plans

d) raise incidents on faults that they have found

e) provide information for risk analysis and quality

improvement 29 Unreachable code would best be found using:

a) code reviews

b) code inspections

c) a coverage tool

d) a test management tool

e) a static analysis tool

30 A tool that supports traceability, recording of incidents or scheduling of tests is called:

a) a dynamic analysis tool

b) a test execution tool

c) a debugging tool

d) a test management tool

e) a configuration management tool

31 What information need not be included in a test incident

report: a) how to fix the fault

b) how to reproduce the fault

c) test environment details

d) severity, priority

e) the actual and expected outcomes

32 Which expression best matches the following characteristics or review processes:

1. led by author

2. undocumented

3. no management participation

4. led by a trained moderator or leader

5. uses entry exit criteria

s) inspection

t) peer review

u) informal review

v) walkthrough

- a) s = 4, t = 3, u = 2 and 5, v = 1
- b) s = 4 and 5, t = 3, u = 2, v = 1
- c) s = 1 and 5, t = 3, u = 2, v = 4
- d) s = 5, t = 4, u = 3, v = 1 and 2
- e) s = 4 and 5, t = 1, u = 2, v = 3

33 Which of the following is NOT part of system testing:

- a) business process-based testing
- b) performance, load and stress testing
- c) requirements-based testing
- d) usability testing
- e) top-down integration testing

34 What statement about expected outcomes is FALSE:

- a) expected outcomes are defined by the software's behaviour
- b) expected outcomes are derived from a specification, not from the code
- c) expected outcomes include outputs to a screen and changes to files and databases
- d) expected outcomes should be predicted before a test is run
- e) expected outcomes may include timing constraints such as response

times 35 The standard that gives definitions of testing terms is:

- a) ISO/IEC 12207
- b) BS7925-1
- c) BS7925-2
- d) ANSI/IEEE 829
- e) ANSI/IEEE 729

36 The cost of fixing a fault:

- a) Is not important
- b) Increases as we move the product towards live use
- c) Decreases as we move the product towards live use
- d) Is more expensive if found in requirements than functional design
- e) Can never be determined

37 Which of the following is NOT included in the Test Plan document of the Test Documentation Standard:

- a) Test items (i.e. software versions)
- b) What is not to be tested
- c) Test environments
- d) Quality plans
- e) Schedules and deadlines

38 Could reviews or inspections be considered part of testing:

- a) No, because they apply to development documentation
- b) No, because they are normally applied before testing
- c) No, because they do not apply to the test documentation
- d) Yes, because both help detect faults and improve quality

- e) Yes, because testing includes all non-constructive activities
- 39 Which of the following is not part of performance testing:
- a) Measuring response time
 - b) Measuring transaction rates
 - c) Recovery testing
 - d) Simulating many users
 - e) Generating many transactions
- 40 Error guessing is best used
- a) As the first approach to deriving test cases b)
After more formal techniques have been applied
 - c) By inexperienced testers
 - d) After the system has gone live
 - e) Only by end users
1. An input field takes the year of birth between 1900 and 2004
The boundary values for testing this field are
- a. 0,1900,2004,2005
 - b. 1900, 2004
 - c. 1899,1900,2004,2005
 - d. 1899, 1900, 1901,2003,2004,2005
2. Which one of the following are non-functional testing methods?
- a. System testing
 - b. Usability testing
 - c. Performance testing
 - d. Both b & c
3. Which of the following tools would be involved in the automation of regression test?
- a. Data tester
 - b. Boundary tester
 - c. Capture/Playback
 - d. Output comparator.
4. Incorrect form of Logic coverage is:
- a. Statement Coverage
 - b. Pole Coverage
 - c. Condition Coverage
 - d. Path Coverage
5. Which of the following is not a quality characteristic listed in ISO 9126 Standard?
- a. Functionality
 - b. Usability
 - c. Supportability
 - d. Maintainability
6. To test a function, the programmer has to write a _____, which calls the function to be tested and passes it test data.
- a. Stub

- b. Driver
 - c. Proxy
 - d. None of the above
7. Boundary value testing
- a. Is the same as equivalence partitioning tests
 - b. Test boundary conditions on, below and above the edges of input and output equivalence classes
 - c. Tests combinations of input circumstances
 - d. Is used in white box testing strategy
8. Pick the best definition of quality
- a. Quality is job one
 - b. Zero defects
 - c. Conformance to requirements
 - d. Work as designed
9. Fault Masking is
- a. Error condition hiding another error condition
 - b. Creating a test case which does not reveal a fault
 - c. Masking a fault by developer
 - d. Masking a fault by a tester
10. One Key reason why developers have difficulty testing their own work is :
- a. Lack of technical documentation
 - b. Lack of test tools on the market for developers
 - c. Lack of training
 - d. Lack of Objectivity
11. During the software development process, at what point can the test process start?
- a. When the code is complete.
 - b. When the design is complete.
 - c. When the software requirements have been approved.
 - d. When the first code module is ready for unit testing
12. In a review meeting a moderator is a person who
- a. Takes minutes of the meeting
 - b. Mediates between people
 - c. Takes telephone calls
 - d. Writes the documents to be reviewed
13. Given the Following program
- ```
IF X <>= Z
THEN Statement 2;
END
```
- McCabe's Cyclomatic Complexity is :
- a. 2
  - b. 3
  - c. 4
  - d. 5

14. How many test cases are necessary to cover all the possible sequences of statements (paths) for the following program fragment? Assume that the two conditions are independent of each other : -

.....

```
if (Condition 1)
then statement 1
else statement 2
fi
if (Condition 2)
then statement 3
fi
```

www.istqb.guru

- .....
- a. 2 Test Cases
  - b. 3 Test Cases
  - c. 4 Test Cases
  - d. Not achievable
15. Acceptance test cases are based on what?
- a. Requirements
  - b. Design
  - c. Code
  - d. Decision table
16. "How much testing is enough?"
- a. This question is impossible to answer
  - b. This question is easy to answer
  - c. The answer depends on the risk for your industry, contract and special requirements
  - d. This answer depends on the maturity of your developers
17. A common test technique during component test is
- a. Statement and branch testing
  - b. Usability testing
  - c. Security testing
  - d. Performance testing
18. Statement Coverage will not check for the following.
- a. Missing Statements
  - b. Unused Branches
  - c. Dead Code
  - d. Unused Statement
19. Independent Verification & Validation is
- a. Done by the Developer
  - b. Done by the Test Engineers
  - c. Done By Management
  - d. Done by an Entity Outside the Project's sphere of

**influence** 20. Code Coverage is used as a measure of what ?

- a. Defects
- b. Trends analysis
- c. Test Effectiveness**
- d. Time Spent Testing

1 We split testing into distinct stages primarily because:

- a) Each test stage has a different purpose.**
- b) It is easier to manage testing in stages.
- c) We can run different tests in different environments.
- d) The more stages we have, the better the testing.

2 Which of the following is likely to benefit most from the use of test tools providing test capture and replay facilities?

- a) Regression testing**
- b) Integration testing
- c) System testing
- d) User acceptance testing

3 Which of the following statements is NOT correct?

- a) A minimal test set that achieves 100% LCSAJ coverage will also achieve 100% branch coverage.
- b) A minimal test set that achieves 100% path coverage will also achieve 100% **statement coverage.**
- c) A minimal test set that achieves 100% path coverage will generally detect more faults than one that achieves 100% statement coverage.
- d) A minimal test set that achieves 100% statement coverage will generally detect more faults than one that achieves 100% branch coverage.

4 Which of the following requirements is testable?

- a) The system shall be user friendly.
- b) The safety-critical parts of the system shall contain 0 faults.
- c) The response time shall be less than one second for the specified design load.**
- d) The system shall be built to be portable.

5 Analyse the following highly simplified procedure:

Ask: "What type of ticket do you require, single or return?"

IF the customer wants 'return'

Ask: "What rate, Standard or Cheap-day?"

IF the customer replies 'Cheap-day'

Say: "That will be £11:20"

ELSE

Say: "That will be £19:50"

ENDIF

ELSE

Say: "That will be £9:75"

ENDIF

Now decide the minimum number of tests that are needed to ensure that all the questions have been asked, all combinations have occurred and

all replies given.

a) 3

b) 4

c) 5

d) 6

6. Error guessing:

a) supplements formal test design techniques.

b) can only be used in component, integration and system testing. c) is only performed in user acceptance testing.

d) is not repeatable and should not be used.

7 Which of the following is NOT true of test coverage criteria?

a) Test coverage criteria can be measured in terms of items exercised by a test suite.

b) A measure of test coverage criteria is the percentage of user requirements covered.

c) A measure of test coverage criteria is the percentage of faults found. d)

Test coverage criteria are often used when specifying test completion criteria. 8 In prioritising what to test, the most important objective is to:

a) find as many faults as possible.

b) test high risk areas.

c) obtain good test coverage.

d) test whatever is easiest to test.

9 Given the following sets of test management terms (v-z), and activity descriptions (1-5), which one of the following best pairs the two sets? v – test control

w – test monitoring

x - test estimation

y - incident management

z - configuration control

1 - calculation of required test resources

2 - maintenance of record of test results

3 - re-allocation of resources when tests overrun

4 - report on deviation from test plan

5 - tracking of anomalous test results

a) v-3,w-2,x-1,y-5,z-4

b) v-2,w-5,x-1,y-4,z-3

c) v-3,w-4,x-1,y-5,z-2

d) v-2,w-1,x-4,y-3,z-5

10 Which one of the following statements about system testing is NOT true? a) System tests are often performed by independent teams.

b) Functional testing is used more than structural testing.

c) Faults found during system tests can be very expensive to fix.

d) End-users should be involved in system tests.

11 Which of the following is false?

a) Incidents should always be fixed.

- b) An incident occurs when expected and actual results differ.
- c) Incidents can be analysed to assist in test process improvement.
- d) An incident can be raised against documentation.

12 Enough testing has been performed when:

- a) time runs out.
- b) the required level of confidence has been achieved.
- c) no more faults are found.
- d) the users won't find any serious faults.

13 Which of the following is NOT true of incidents?

- a) Incident resolution is the responsibility of the author of the software under test.
- b) Incidents may be raised against user requirements.
- c) Incidents require investigation and/or correction.
- d) Incidents are raised when expected and actual results differ.

14 Which of the following is not described in a unit test standard?

- a) syntax testing
- b) equivalence partitioning
- c) stress testing
- d) modified condition/decision coverage

15 Which of the following is false?

- a) In a system two different failures may have different severities.
- b) A system is necessarily more reliable after debugging for the removal of a fault.
- c) A fault need not affect the reliability of a system.
- d) Undetected errors may lead to faults and eventually to incorrect behaviour.

16 Which one of the following statements, about capture-replay tools, is NOT correct?

- a) They are used to support multi-user testing.
- b) They are used to capture and animate user requirements.
- c) They are the most frequently purchased types of CAST tool.
- d) They capture aspects of user behaviour.

17 How would you estimate the amount of re-testing likely to be required?

- a) Metrics from previous similar projects
- b) Discussions with the development team
- c) Time allocated for regression testing
- d) a & b

18 Which of the following is true of the V-model?

- a) It states that modules are tested against user requirements.
- b) It only models the testing phase.
- c) It specifies the test techniques to be used.
- d) It includes the verification of designs.

19 The oracle assumption:

- a) is that there is some existing system against which test output may be checked.
- b) is that the tester can routinely identify the correct outcome of a

test. c) is that the tester knows everything about the software under test. d) is that the tests are reviewed by experienced testers.

20 Which of the following characterises the cost of faults?

a) They are cheapest to find in the early development phases and the most expensive to fix in the latest test phases.

b) They are easiest to find during system testing but the most expensive to fix then.

c) Faults are cheapest to find in the early development phases but the most expensive to fix then.

d) Although faults are most expensive to find during early development phases, they are cheapest to fix then.

21 Which of the following should NOT normally be an objective for a test? a) To find faults in the software.

b) To assess whether the software is ready for release.

c) To demonstrate that the software doesn't work.

d) To prove that the software is correct.

22 Which of the following is a form of functional testing?

a) Boundary value analysis

b) Usability testing

c) Performance testing

d) Security testing

23 Which of the following would NOT normally form part of a test plan? a) Features to be tested

b) Incident reports

c) Risks

d) Schedule

24 Which of these activities provides the biggest potential cost saving from the use of CAST? Computer Aided SW Testing

a) Test management

b) Test design

c) Test execution

d) Test planning

25 Which of the following is NOT a white box technique?

a) Statement testing

b) Path testing

c) Data flow testing

d) State transition testing

26 Data flow analysis studies:

a) possible communications bottlenecks in a program.

b) the rate of change of data values as a program executes.

c) the use of data on paths through the code.

d) the intrinsic complexity of the code.

27 In a system designed to work out the tax to be paid:

An employee has £4000 of salary tax free. The next £1500 is taxed

at 10% The next £28000 is taxed at 22%

Any further amount is taxed at 40%

To the nearest whole pound, which of these is a valid Boundary Value Analysis test case?

- a) £1500
- b) £32001
- c) £33501
- d) £28000

28 An important benefit of code inspections is that they:

- a) enable the code to be tested before the execution environment is ready.
- b) can be performed by the person who wrote the code.
- c) can be performed by inexperienced staff.
- d) are cheap to perform.

29 Which of the following is the best source of Expected Outcomes for User Acceptance Test scripts?

- a) Actual results
- b) Program specification
- c) User requirements
- d) System specification

30 What is the main difference between a walkthrough and an inspection? a) An inspection is lead by the author, whilst a walkthrough is lead by a trained moderator.

- b) An inspection has a trained leader, whilst a walkthrough has no leader.
- c) Authors are not present during inspections, whilst they are during walkthroughs.
- d) A walkthrough is lead by the author, whilst an inspection is lead by a trained moderator.

31 Which one of the following describes the major benefit of verification early in the life cycle?

- a) It allows the identification of changes in user requirements.
- b) It facilitates timely set up of the test environment.
- c) It reduces defect multiplication.
- d) It allows testers to become involved early in the project.

32 Integration testing in the small:

- a) tests the individual components that have been developed.
- b) tests interactions between modules or subsystems.
- c) only uses components that form part of the live system.
- d) tests interfaces to other systems.

33 Static analysis is best described as:

- a) the analysis of batch programs.
- b) the reviewing of test plans.
- c) the analysis of program code.
- d) the use of black box testing.

34 Alpha testing is:

- a) post-release testing by end user representatives at the developer's

site. b) the first testing that is performed.

c) pre-release testing by end user representatives at the developer's site. d) pre-release testing by end user representatives at their sites.

35 A failure is:

a) found in the software; the result of an error.

b) departure from specified behaviour.

c) an incorrect step, process or data definition in a computer program. d) a human action that produces an incorrect result.

36 In a system designed to work out the tax to be paid:

An employee has £4000 of salary tax free. The next £1500 is taxed at 10% The next £28000 is taxed at 22%

Any further amount is taxed at 40%

Which of these groups of numbers would fall into the same equivalence class? a) £4800; £14000; £28000

b) £5200; £5500; £28000

c) £28001; £32000; £35000

d) £5800; £28000; £32000

37 The most important thing about early test design is that it: a) makes test preparation easier.

b) means inspections are not required.

c) can prevent fault multiplication.

d) will find all faults.

38 Which of the following statements about reviews is true?

a) Reviews cannot be performed on user requirements specifications. b) Reviews are the least effective way of testing code.

c) Reviews are unlikely to find faults in test plans.

d) Reviews should be performed on specifications, code, and test plans.

39 Test cases are designed during:

a) test recording.

b) test planning.

c) test configuration.

d) test specification.

40 A configuration management system would NOT normally provide: a) linkage of customer requirements to version numbers.

b) facilities to compare test results with expected results.

c) the precise differences in versions of software component source code. d) restricted access to the source code library.

1. Software testing activities should start

a. as soon as the code is written

b. during the design stage

c. when the requirements have been formally documented

d. as soon as possible in the development life cycle

2. Faults found by users are due to:
  - a. Poor quality software
  - b. Poor software and poor testing
  - c. bad luck
  - d. insufficient time for testing
3. What is the main reason for testing software before releasing it?
  - a. to show that system will work after release
  - b. to decide when the software is of sufficient quality to release
  - c. to find as many bugs as possible before release
  - d. to give information for a risk based decision about release
4. Which of the following statements is not true
  - a. performance testing can be done during unit testing as well as during the testing of whole system
  - b. The acceptance test does not necessarily include a regression test
  - c. Verification activities should not involve testers (reviews, inspections etc)
  - d. Test environments should be as similar to production environments as possible
5. When reporting faults found to developers, testers should be:
  - a. as polite, constructive and helpful as possible
  - b. firm about insisting that a bug is not a "feature" if it should be fixed
  - c. diplomatic, sensitive to the way they may react to criticism
  - d. All of the above
6. In which order should tests be run?
  - a. the most important tests first
  - b. the most difficult tests first (to allow maximum time for fixing)
  - c. the easiest tests first (to give initial confidence)
  - d. the order they are thought of
7. The later in the development life cycle a fault is discovered, the more expensive it is to fix. Why?
  - a. the documentation is poor, so it takes longer to find out what the software is doing.
  - b. wages are rising
  - c. the fault has been built into more documentation, code, tests, etc
  - d. none of the above
8. Which is not true - The black box tester
  - a. should be able to understand a functional specification or requirements document
  - b. should be able to understand the source code.
  - c. is highly motivated to find faults
  - d. is creative to find the system's weaknesses
9. A test design technique is
  - a. a process for selecting test cases
  - b. a process for determining expected outputs
  - c. a way to measure the quality of software
  - d. a way to measure in a test plan what has to be done

10. Testware(test cases, test dataset)
- a. needs configuration management just like requirements, design and code
  - b. should be newly constructed for each new version of the software
  - c. is needed only until the software is released into production or use
  - d. does not need to be documented and commented, as it does not form part of the released software system
11. An incident logging system
- a only records defects
  - b is of limited value
  - c is a valuable source of project information during testing if it contains all incidents
  - d. should be used only by the test team.
12. Increasing the quality of the software, by better development methods, will affect the time needed for testing (the test phases) by:
- a. reducing test time
  - b. no change
  - c. increasing test time
  - d. can't say
13. Coverage measurement
- a. is nothing to do with testing
  - b. is a partial measure of test thoroughness
  - c. branch coverage should be mandatory for all software
  - d. can only be applied at unit or module testing, not at system testing
14. When should you stop testing?
- a. when time for testing has run out.
  - b. when all planned tests have been run
  - c. when the test completion criteria have been met
  - d. when no faults have been found by the tests run
15. Which of the following is true?
- a. Component testing should be black box, system testing should be white box.
  - b. if u find a lot of bugs in testing, you should not be very confident about the quality of software
  - c. the fewer bugs you find,the better your testing was
  - d. the more tests you run, the more bugs you will find.
16. What is the important criterion in deciding what testing technique to use?
- a. how well you know a particular technique
  - b. the objective of the test
  - c. how appropriate the technique is for testing the application
  - d. whether there is a tool to support the technique
17. If the pseudocode below were a programming language ,how many tests are required to achieve 100% statement coverage?
- ```
If x=3 then
    Display_messageX;
    If y=2 then
```

```
        Display_messageY;  
Else  
        Display_messageZ;  
Else  
        Display_messageZ;
```

- a. 1
- b. 2
- c. 3
- d. 4

18. Using the same code example as question 17, how many tests are required to achieve 100% branch/decision coverage?

- a. 1
- b. 2
- c. 3
- d. 4

19 Which of the following is NOT a type of non-functional test? a. State-Transition

- b. Usability
- c. Performance
- d. Security

20. Which of the following tools would you use to detect a memory leak? a. State analysis

- b. Coverage analysis
- c. Dynamic analysis
- d. Memory analysis

21. Which of the following is NOT a standard related to testing? a. IEEE829

- b. IEEE610
- c. BS7925-1
- d. BS7925-2

22. which of the following is the component test standard?

- a. IEEE 829
- b. IEEE 610
- c. BS7925-1
- d. BS7925-2

23 which of the following statements are true?

- a. Faults in program specifications are the most expensive to fix.
- b. Faults in code are the most expensive to fix.
- c. Faults in requirements are the most expensive to fix.
- d. Faults in designs are the most expensive to fix.

24. Which of the following is not the integration strategy?

- a. Design based
- b. Big-bang
- c. Bottom-up

- d. Top-down
25. Which of the following is a black box design technique?
- a. statement testing
 - b. equivalence partitioning
 - c. error- guessing
 - d. usability testing
26. A program with high cyclometric complexity is almost likely to be:
- a. Large
 - b. Small
 - c. Difficult to write
 - d. Difficult to test
27. Which of the following is a static test?
- a. code inspection
 - b. coverage analysis
 - c. usability assessment
 - d. installation test
28. Which of the following is the odd one out?
- a. white box
 - b. glass box
 - c. structural
 - d. functional
29. A program validates a numeric field as follows:
values less than 10 are rejected, values between 10 and 21 are accepted,
values greater than or equal to 22 are rejected
which of the following input values cover all of the equivalence
partitions?
- a. 10,11,21
 - b. 3,20,21
 - c. 3,10,22
 - d. 10,21,22
30. Using the same specifications as question 29, which of the following covers
the MOST boundary values?
- a. 9,10,11,22
 - b. 9,10,21,22
 - c. 10,11,21,22
 - d. 10,11,20,21
- 1 When what is visible to end-users is a deviation from the specific or
expected behavior, this is called:
- a) an error
 - b) a fault
 - c) a failure
 - d) a defect
 - e) a mistake
- 2 Regression testing should be performed:
- v) every week

- w) after the software has changed
 - x) as often as possible
 - y) when the environment has changed
 - z) when the project manager says
 - a) v & w are true, x – z are false
 - b) w, x & y are true, v & z are false
 - c) w & y are true, v, x & z are false
 - d) w is true, v, x y and z are false
 - e) all of the above are true
- 3 IEEE 829 test plan documentation standard contains all of the following except: a) test items
- b) test deliverables
 - c) test tasks
 - d) test environment
 - e) test specification
- 4 Testing should be stopped when:
- a) all the planned tests have been run
 - b) time has run out
 - c) all faults have been fixed correctly
 - d) both a) and c)
 - e) it depends on the risks for the system being tested
- 5 Order numbers on a stock control system can range between 10000 and 99999 inclusive. Which of the following inputs might be a result of designing tests for only valid equivalence classes and valid boundaries:
- a) 1000, 5000, 99999
 - b) 9999, 50000, 100000
 - c) 10000, 50000, 99999
 - d) 10000, 99999
 - e) 9999, 10000, 50000, 99999, 10000
- 6 Consider the following statements about early test design:
- i. early test design can prevent fault multiplication
 - ii. faults found during early test design are more expensive to fix
 - iii. early test design can find faults
 - iv. early test design can cause changes to the requirements
 - v. early test design takes more effort
 - a) i, iii & iv are true. ii & v are false
 - b) iii is true, i, ii, iv & v are false
 - c) iii & iv are true. i, ii & v are false
 - d) i, iii, iv & v are true, ii is false
 - e) i & iii are true, ii, iv & v are false
- 7 Non-functional system testing includes:
- a) testing to see where the system does not function properly
 - b) testing quality attributes of the system including performance and usability
 - c) testing a system feature using only the software required for that action
 - d) testing a system feature using only the software required for

that function e) testing for functions that should not exist

8 Which of the following is NOT part of configuration management: a) status accounting of configuration items

b) auditing conformance to ISO9001

c) identification of test versions

d) record of changes to documentation over time

e) controlled library access

9 Which of the following is the main purpose of the integration strategy for integration testing in the small?

a) to ensure that all of the small modules are tested adequately

b) to ensure that the system interfaces to other systems and networks

c) to specify which modules to combine when and how many at once

d) to ensure that the integration testing can be performed by a small team

e) to specify how the software should be divided into modules

10 What is the purpose of test completion criteria in a test plan:

a) to know when a specific test has finished its execution

b) to ensure that the test case specification is complete

c) to set the criteria used in generating test inputs

d) to know when test planning is complete

e) to plan when to stop testing

11 Consider the following statements

i. an incident may be closed without being fixed

ii. incidents may not be raised against documentation

iii. the final stage of incident tracking is fixing

iv. the incident record does not include information on test environments

v. incidents should be raised when someone other than the author of the software performs the test

a) ii and v are true, i, iii and iv are false

b) i and v are true, ii, iii and iv are false

c) i, iv and v are true, ii and iii are false

d) i and ii are true, iii, iv and v are false

e) i is true, ii, iii, iv and v are false

12 Given the following code, which is true about the minimum number of test cases required for full statement and branch coverage:

```
Read P
```

```
Read Q
```

```
IF P+Q > 100 THEN
```

```
Print "Large"
```

```
ENDIF
```

```
If P > 50 THEN
```

```
Print "P Large"
```

```
ENDIF
```

a) 1 test for statement coverage, 3 for branch coverage

b) 1 test for statement coverage, 2 for branch coverage

- c) 1 test for statement coverage, 1 for branch coverage
- d) 2 tests for statement coverage, 3 for branch coverage
- e) 2 tests for statement coverage, 2 for branch coverage

13 Given the following:

Switch PC on

Start "outlook"

IF outlook appears THEN

Send an email

Close outlook

- a) 1 test for statement coverage, 1 for branch coverage
- b) 1 test for statement coverage, 2 for branch coverage
- c) 1 test for statement coverage. 3 for branch coverage
- d) 2 tests for statement coverage, 2 for branch coverage
- e) 2 tests for statement coverage, 3 for branch coverage

14 Given the following code, which is true:

IF A > B THEN

C = A – B

ELSE

C = A + B

ENDIF

Read D

IF C = D Then

Print "Error"

ENDIF

- a) 1 test for statement coverage, 3 for branch coverage
- b) 2 tests for statement coverage, 2 for branch coverage
- c) 2 tests for statement coverage. 3 for branch coverage
- d) 3 tests for statement coverage, 3 for branch coverage
- e) 3 tests for statement coverage, 2 for branch coverage

15 Consider the following:

Pick up and read the newspaper

Look at what is on television

If there is a program that you are interested in watching then switch the the television on and watch the program

Otherwise

Continue reading the newspaper

If there is a crossword in the newspaper then try and complete the

crossword a) SC = 1 and DC = 1

b) SC = 1 and DC = 2

c) SC = 1 and DC = 3

d) SC = 2 and DC = 2

e) SC = 2 and DC = 3

16 The place to start if you want a (new) test tool is:

a) Attend a tool exhibition

b) Invite a vendor to give a demo

c) Analyse your needs and requirements

d) Find out what your budget would be for the tool

e) Search the internet

17 When a new testing tool is purchased, it should be used first

by: a) A small team to establish the best way to use the

tool b) Everyone who may eventually have some use for the tool

c) The independent testing team

d) The managers to see what projects it should be

used in e) The vendor contractor to write the initial scripts

18 What can static analysis NOT find?

a) The use of a variable before it has been defined

b) Unreachable ("dead") code

c) Whether the value stored in a variable is correct

d) The re-definition of a variable before it has been used e) Array bound violations

19 Which of the following is NOT a black box technique:

a) Equivalence partitioning

b) State transition testing

c) LCSAJ

d) Syntax testing

e) Boundary value analysis

20 Beta testing is:

a) Performed by customers at their own site

b) Performed by customers at their software developer's site c) Performed by an independent test team

d) Useful to test bespoke software

e) Performed as early as possible in the lifecycle

21 Given the following types of tool, which tools would typically be used by developers and which by an independent test team:

i. static analysis

ii. performance testing

iii. test management

iv. dynamic analysis

v. test running

vi. test data preparation

a) developers would typically use i, iv and vi; test team ii, iii and v b) developers would typically use i and iv; test team ii,

iii, v and vi c) developers would typically use i, ii, iii and iv;

test team v and vi d) developers would typically use ii, iv and

vi; test team i, ii and v e) developers would typically use i, iii,

iv and v; test team ii and vi

22 The main focus of acceptance testing is:

- a) finding faults in the system
- b) ensuring that the system is acceptable to all users
- c) testing the system with other systems
- d) testing for a business perspective
- e) testing by an independent test team

23 Which of the following statements about the component testing standard is false:

- a) black box design techniques all have an associated measurement technique
 - b) white box design techniques all have an associated measurement technique
 - c) cyclomatic complexity is not a test measurement technique
 - d) black box measurement techniques all have an associated test design technique
 - e) white box measurement techniques all have an associated test design technique
- 24 Which of the following statements is NOT true:

- a) inspection is the most formal review process
 - b) inspections should be led by a trained leader
 - c) managers can perform inspections on management documents
 - d) inspection is appropriate even when there are no written documents
 - e) inspection compares documents with predecessor (source) documents
- 25 A typical commercial test execution tool would be able to perform all of the following EXCEPT:

- a) generating expected outputs
- b) replaying inputs according to a programmed script
- c) comparison of expected outcomes with actual outcomes
- d) recording test inputs
- e) reading test values from a data file

26 The difference between re-testing and regression testing is

- a) re-testing is running a test again; regression testing looks for unexpected side effects
- b) re-testing looks for unexpected side effects; regression testing is repeating those tests
- c) re-testing is done after faults are fixed; regression testing is done earlier
- d) re-testing uses different environments, regression testing uses the same environment
- e) re-testing is done by developers, regression testing is done by independent testers

27 Expected results are:

- a) only important in system testing
- b) only used in component testing
- c) never specified in advance
- d) most useful when specified in advance
- e) derived from the code

28 Test managers should not:

- a) report on deviations from the project plan

- b) sign the system off for release
- c) re-allocate resource to meet original plans
- d) raise incidents on faults that they have found
- e) provide information for risk analysis and quality

improvement 29 Unreachable code would best be found using:

- a) code reviews
- b) code inspections
- c) a coverage tool
- d) a test management tool
- e) a static analysis tool

30 A tool that supports traceability, recording of incidents or scheduling of tests is called:

- a) a dynamic analysis tool
- b) a test execution tool
- c) a debugging tool
- d) a test management tool
- e) a configuration management tool

31 What information need not be included in a test incident

- report:
- a) how to fix the fault
 - b) how to reproduce the fault
 - c) test environment details
 - d) severity, priority
 - e) the actual and expected outcomes

32 Which expression best matches the following characteristics or review processes:

1. led by author
2. undocumented
3. no management participation
4. led by a trained moderator or leader
5. uses entry exit criteria

s) inspection

t) peer review

u) informal review

v) walkthrough

- a) s = 4, t = 3, u = 2 and 5, v = 1
- b) s = 4 and 5, t = 3, u = 2, v = 1
- c) s = 1 and 5, t = 3, u = 2, v = 4
- d) s = 5, t = 4, u = 3, v = 1 and 2
- e) s = 4 and 5, t = 1, u = 2, v = 3

33 Which of the following is NOT part of system testing:

- a) business process-based testing
- b) performance, load and stress testing
- c) requirements-based testing
- d) usability testing

e) top-down integration testing

34 What statement about expected outcomes is FALSE:

- a) expected outcomes are defined by the software's behaviour b) expected outcomes are derived from a specification, not from the code
- c) expected outcomes include outputs to a screen and changes to files and databases
- d) expected outcomes should be predicted before a test is run
- e) expected outcomes may include timing constraints such as response times

35 The standard that gives definitions of testing terms is:

- a) ISO/IEC 12207
- b) BS7925-1
- c) BS7925-2
- d) ANSI/IEEE 829
- e) ANSI/IEEE 729

36 The cost of fixing a fault:

- a) Is not important
- b) Increases as we move the product towards live use
- c) Decreases as we move the product towards live use
- d) Is more expensive if found in requirements than functional design
- e) Can never be determined

37 Which of the following is NOT included in the Test Plan document of the Test Documentation Standard:

- a) Test items (i.e. software versions)
- b) What is not to be tested
- c) Test environments
- d) Quality plans
- e) Schedules and deadlines

38 Could reviews or inspections be considered part of testing:

- a) No, because they apply to development documentation
- b) No, because they are normally applied before testing
- c) No, because they do not apply to the test documentation
- d) Yes, because both help detect faults and improve quality
- e) Yes, because testing includes all non-constructive activities

39 Which of the following is not part of performance testing:

- a) Measuring response time
- b) Measuring transaction rates
- c) Recovery testing
- d) Simulating many users
- e) Generating many transactions

40 Error guessing is best used

- a) As the first approach to deriving test cases
- b) After more formal techniques have been applied

- c) By inexperienced testers
 - d) After the system has gone live
 - e) Only by end users
1. COTS is known as
 - A. Commercial off the shelf software
 - B. Compliance of the software
 - C. Change control of the software
 - D. Capable off the shelf software
 2. From the below given choices, which one is the 'Confidence testing'
 - A. Sanity testing
 - B. System testing
 - C. Smoke testing
 - D. Regression testing
 3. 'Defect Density' calculated in terms of
 - A. The number of defects identified in a component or system divided by the size of the component or the system
 - B. The number of defects found by a test phase divided by the number found by that test phase and any other means afterwards
 - C. The number of defects identified in the component or system divided by the number of defects found by a test phase
 - D. The number of defects found by a test phase divided by the number found by the size of the system
 4. 'Be bugging' is known as
 - A. Preventing the defects by inspection
 - B. Fixing the defects by debugging
 - C. Adding known defects by seeding
 - D. A process of fixing the defects by tester
 5. An expert based test estimation is also known as
 - A. Narrow band Delphi
 - B. Wide band Delphi
 - C. Bespoke Delphi
 - D. Robust Delphi
 6. When testing a grade calculation system, a tester determines that all scores from 90 to 100 will yield a grade of A, but scores below 90 will not. This analysis is known as:
 - A. Equivalence partitioning
 - B. Boundary value analysis
 - C. Decision table
 - D. Hybrid analysis
 7. All of the following might be done during unit testing except
 - A. Desk check
 - B. Manual support testing
 - C. Walkthrough
 - D. Compiler based testing

8. Find the Min number of tests to ensure that each statement is executed at least once

- A. 5
- B. 6
- C. 4
- D. 8

9. Which of the following characteristics is primarily associated with software reusability?

- A. The extent to which the software can be used in other applications
- B. The extent to which the software can be used by many different users
- C. The capability of the software to be moved to a different platform
- D. The capability of one system to be coupled with another system

10. Which of the following software change management activities is most vital to assessing the impact of proposed software modifications?

- A. Baseline identification
- B. Configuration auditing
- C. Change control
- D. Version control

11. Which of the following statements is true about a software verification and validation program?

- I. It strives to ensure that quality is built into software.
- II. It provides management with insights into the state of a software project.
- III. It ensures that alpha, beta, and system tests are performed.
- IV. It is executed in parallel with software development activities.

- A. I, II&III
- B. II, III&IV
- C. I, II&IV
- D. I, III&IV

12. Which of the following is a requirement of an effective software environment?

- I. Ease of use
- II. Capacity for incremental implementation
- III. Capability of evolving with the needs of a project
- IV. Inclusion of advanced tools

- A. I, II & III
- B. I, II & IV
- C. II, III&IV
- D. I, III&IV

13. A test manager wants to use the resources available for the automated testing of a web application. The best choice is

- A. Test automater, web specialist, DBA, test lead
- B. Tester, test automater, web specialist, DBA

- C. Tester, test lead, test automater, DBA
 - D. Tester, web specialist, test lead, test automater
14. A project manager has been transferred to a major software development project that is in the implementation phase. The highest priority for this project manager should be to
- B. Establish a relationship with the customer
 - C. Learn the project objectives and the existing project plan
 - D. Modify the project's organizational structure to meet the manager's management style
 - E. Ensure that the project proceeds at its current pace
15. Change X requires a higher level of authority than Change Y in which of the following pairs?
- | | |
|---|------------------------------------|
| Change X | Change Y |
| A. Code in development | Code in production |
| B. Specifications during requirements analysis | Specifications during systems test |
| C. Documents requested by the technical development group | Documents requested by customers |
| D. A product distributed to several sites | A product with a single user |
16. Which of the following functions is typically supported by a software quality information system?
- I. Record keeping
 - II. System design
 - III. Evaluation scheduling
 - IV. Error reporting
- A. I, II&III
 - B. II, III & IV
 - C. I, III & IV
 - D. I, II & IV
17. During the testing of a module tester 'X' finds a bug and assigned it to developer. But developer rejects the same, saying that it's not a bug. What 'X' should do?
- A. Report the issue to the test manager and try to settle with the developer.
 - B. Retest the module and confirm the bug
 - C. Assign the same bug to another developer
 - D. Send to the detailed information of the bug encountered and check the reproducibility
18. The primary goal of comparing a user manual with the actual behavior of the running program during system testing is to
- A. Find bugs in the program

B. Check the technical accuracy of the document

C. Ensure the ease of use of the document

D. Ensure that the program is the latest version

19. A type of integration testing in which software elements, hardware elements, or both are combined all at once into a component or an overall system, rather than in stages.

A. System Testing

B. Big-Bang Testing

C. Integration Testing

D. Unit Testing

20. In practice, which Life Cycle model may have more, fewer or different levels of development and testing, depending on the project and the software product. For example, there may be component integration testing after component testing, and system integration testing after system testing.

A. Water Fall Model

B. V-Model

C. Spiral Model

D. RAD Model

21. Which technique can be used to achieve input and output coverage? It can be applied to human input, input via interfaces to a system, or interface parameters in integration testing.

A. Error Guessing

B. Boundary Value Analysis

C. Decision Table testing

D. Equivalence partitioning

22. There is one application, which runs on a single terminal. There is another application that works on multiple terminals. What are the test techniques you will use on the second application that you would not do on the first application? A. Integrity, Response time

B. Concurrency test, Scalability

C. Update & Rollback, Response time

D. Concurrency test, Integrity

23. You are the test manager and you are about to start the system testing. The developer team says that due to change in requirements they will be able to deliver the system to you for testing 5 working days after the due date. You cannot change the resources (work hours, test tools, etc.) What steps will you take to be able to finish the testing in time. (

A. Tell to the development team to deliver the system in time so that testing activity will be finished in time.

B. Extend the testing plan, so that you can accommodate the slip going to occur C. Rank the functionality as per risk and concentrate more on critical functionality testing

D. Add more resources so that the slippage should be avoided

24. Item transmittal report is also known as
- A. Incident report
 - B. Release note**
 - C. Review report
 - D. Audit report
25. Testing of software used to convert data from existing systems for use in replacement systems
- A. Data driven testing
 - B. Migration testing**
 - C. Configuration testing
 - D. Back to back testing
26. Big bang approach is related to
- A. Regression testing
 - B. Inter system testing
 - C. Re-testing
 - D. Integration testing**
27. Cause effect graphing is related to the standard
- A. BS7799
 - B. BS 7925/2**
 - C. ISO/IEC 926/1
 - D. ISO/IEC 2382/1
28. "The tracing of requirements for a test level through the layers of a test documentation" done by
- A. Horizontal traceability**
 - B. Depth traceability
 - C. Vertical traceability
 - D. Horizontal & Vertical traceabilities
29. A test harness is a
- A. A high level document describing the principles, approach and major objectives of the organization regarding testing
 - B. A distance set of test activities collected into a manageable phase of a project**
 - C. A test environment comprised of stubs and drives needed to conduct a test
 - D. A set of several test cases for a component or system under test
30. You are a tester for testing a large system. The system data model is very large with many attributes and there are a lot of inter dependencies with in the fields. What steps would you use to test the system and also what are the efforts of the test you have taken on the test plan
- A. Improve super vision, More reviews of artifacts or program means stage** containment of the defects.
 - B. Extend the test plan so that you can test all the inter dependencies
 - C. Divide the large system in to small modules and test the

functionality D. Test the interdependencies first, after that check the system as a whole

31. Change request should be submitted through development or program management. A change request must be written and should include the following criteria.

- I. Definition of the change
- II. Documentation to be updated
- III. Name of the tester or developer
- IV. Dependencies of the change request.

- A. I, III and IV
- B. I, II and III
- C. II, III and IV
- D. I, II and IV

32. 'Entry criteria' should address questions such as

- I. Are the necessary documentation, design and requirements information available that will allow testers to operate the system and judge correct behavior.
- II. Is the test environment-lab, hardware, software and system administration support ready?
- III. Those conditions and situations that must prevail in the testing process to allow testing to continue effectively and efficiently.
- IV. Are the supporting utilities, accessories and prerequisites available in forms that testers can use

- A. I, II and IV
- B. I, II and III
- C. I, II, III and IV
- D. II, III and IV.

33. "This life cycle model is basically driven by schedule and budget risks"
This statement is best suited for

- A. Water fall model
- B. Spiral model
- C. Incremental model
- D. V-Model

34. The bug tracking system will need to capture these phases for each bug. I. Phase injected

- II. Phase detected
- III. Phase fixed
- IV. Phase removed

- A. I, II and III
- B. I, II and IV
- C. II, III and IV
- D. I, III and IV

35. One of the more daunting challenges of managing a test project is that so

many dependencies converge at test execution. One missing configuration file or hardware device can render all your test results meaning less. You can end up with an entire platoon of testers sitting around for days.

Who is responsible for this incident?

- A. Test managers faults only
- B. Test lead faults only
- C. Test manager and project manager faults
- D. Testers faults only

36. System test can begin when?

- I. The test team completes a three day smoke test and reports on the results to the system test phase entry meeting
- II. The development team provides software to the test team 3 business days prior to starting of the system testing
- III. All components are under formal, automated configuration and release management control

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II and III

37. Test charters are used in _____ testing

- A. Exploratory testing
- B. Usability testing
- C. Component testing
- D. Maintainability testing